

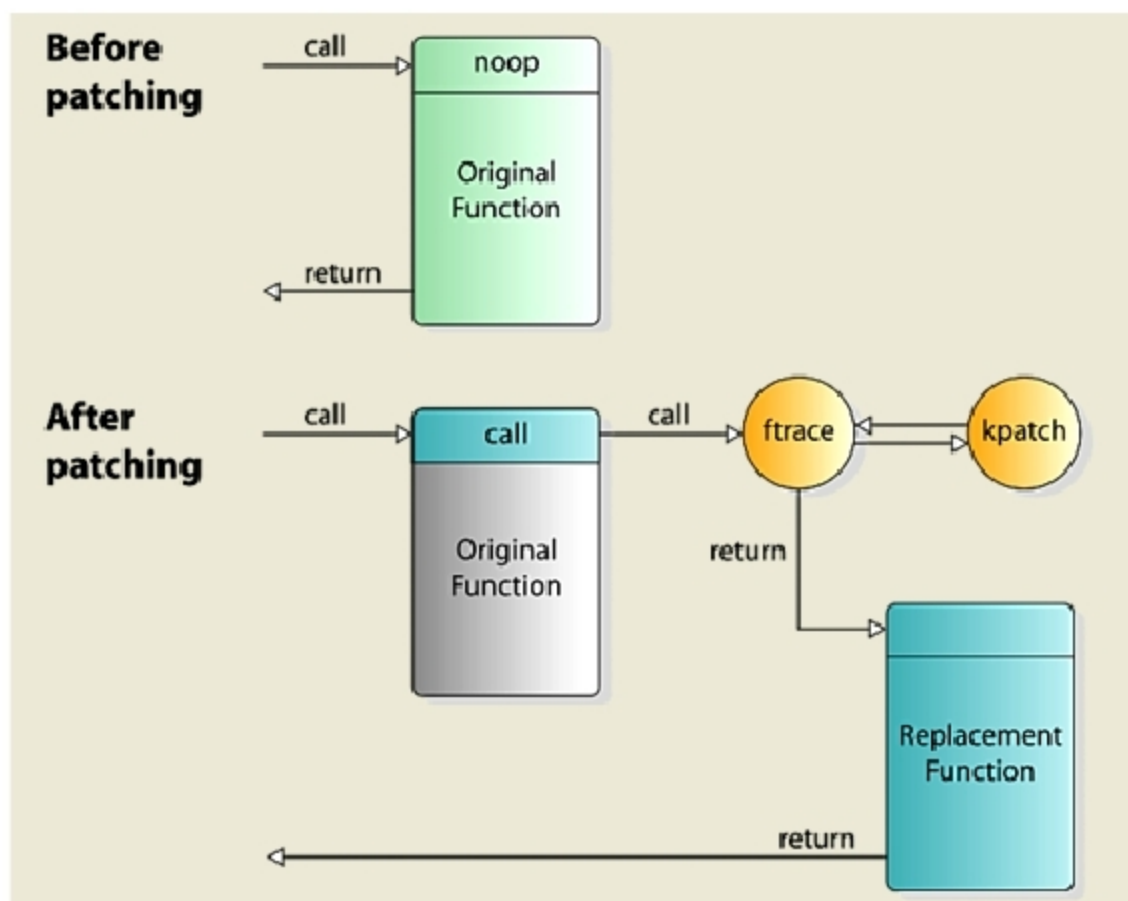


Figure 2: How live patching works

```
File Edit Tabs Help
kousik@kousik:~$ cat /boot/config-$(uname -r) | grep LIVEPATCH
CONFIG_HAVE_LIVEPATCH=y
CONFIG_LIVEPATCH=y
kousik@kousik:~$
```

Figure 3: Live patch enable status

```
kousik@kousik:~$ canonical-livepatch status
2019/05/31 07:33:25.261250 system key.go:124: cannot determine nfs usage in gene
rateSystemKey: cannot parse /etc/fstab: expected between 3 and 6 fields, found 7
client-version: 9.3.0
architecture: x86_64
cpu-model: Intel(R) Core(TM) i3-3110M CPU @ 2.40GHz
last-check: 2019-05-31T07:01:20+05:30
boot-time: 2019-05-31T04:59:20+05:30
uptime: 2h34m5s
status:
- kernel: 4.15.0-51.55-generic
  running: true
  livepatch:
    checkState: checked
    patchState: nothing-to-apply
    version: ""
    fixes: ""
kousik@kousik:~$
```

Figure 4: Running system live patch status

Patches must be disabled in exactly the reverse order in which they were enabled. It makes solving the problem and the implementation much easier.

Unregistration: Disabled patches might be unregistered by calling `klp_unregister_patch()`. This can be done only when the patch is disabled and the code is no longer used. It must be called before the live patch module gets unloaded.

Limitations

Live patching has several limitations. Some of them are:

- Traceable functions only can be patched.
- Kretprobes using the Ftrace framework conflict with the patched functions.
- Kprobes in the original function are ignored when the code is redirected to the new implementation.

OS variants of kernel live patching are:

- KernelCare
- Kexec
- Kgraft
- Kpatch
- Ksplice

Use case of the Ubuntu live patch

Canonical Livepatch Service: This applies critical kernel patches without rebooting.

- Fixes are applied automatically, without restarting your system.
- It reduces downtime, keeping your Ubuntu LTS systems secure and compliant.
- The service is included as part of all Ubuntu Advantage for Infrastructure support packages.
- It is free for three machines.

All you need is an Ubuntu One account.

To check if the live patch feature is enabled or not, run the following command:

```
$cat /boot/config-$(uname -r) | grep
LIVEPATCH
```

To check the current live patch status, run the command given below:

```
$canonical-livepatch status
```



References

- [1] <https://www.kernel.org/doc/Documentation/livepatch/livepatch.txt>
- [2] <https://pdos.csail.mit.edu/papers/ksplice/eurosys.pdf>
- [3] https://en.wikipedia.org/wiki/Dynamic_software Updating
- [4] <https://www.howtoforge.com/history-of-linux-kernel-live-patching/>
- [5] <https://www.ubuntu.com/livepatch>
- [6] <https://linux-audit.com/livepatch-linux-kernel-updates-without-rebooting/>
- [7] <https://lwn.net/Articles/685044/>

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